

April 12, 2022

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50312383

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on March 29, 2022. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Biosolids
Pace Project No.: 50312383

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130
USDA Soil Permit #: P330-19-00257

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SAMPLE SUMMARY

Project: Biosolids

Pace Project No.: 50312383

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50312383001	Biosolids Digester #4	Solid	03/28/22 13:30	03/29/22 08:20

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SAMPLE ANALYTE COUNT

Project: Biosolids

Pace Project No.: 50312383

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50312383001	Biosolids Digester #4	EPA 6020	CAW	10	PASI-I
		SM 2540G	ADT	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50312383

Sample: Biosolids Digester #4 **Lab ID: 50312383001** Collected: 03/28/22 13:30 Received: 03/29/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
6020 MET ICPMS								
Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Pace Analytical Services - Indianapolis								
Arsenic	29.2	mg/kg	1.2	1	04/11/22 12:59	04/12/22 14:46	7440-38-2	
Cadmium	4.6	mg/kg	0.62	1	04/11/22 12:59	04/12/22 14:46	7440-43-9	
Chromium	175	mg/kg	2.5	1	04/11/22 12:59	04/12/22 14:46	7440-47-3	
Copper	1550	mg/kg	24.7	20	04/11/22 12:59	04/12/22 14:27	7440-50-8	
Lead	68.2	mg/kg	6.2	5	04/11/22 12:59	04/12/22 14:32	7439-92-1	
Molybdenum	37.3	mg/kg	1.2	1	04/11/22 12:59	04/12/22 14:46	7439-98-7	N2
Nickel	36.2	mg/kg	0.62	1	04/11/22 12:59	04/12/22 14:46	7440-02-0	
Selenium	8.2	mg/kg	1.2	1	04/11/22 12:59	04/12/22 14:46	7782-49-2	
Silver	6.3	mg/kg	0.62	1	04/11/22 12:59	04/12/22 14:46	7440-22-4	
Zinc	3010	mg/kg	124	20	04/11/22 12:59	04/12/22 14:27	7440-66-6	
Percent Moisture								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Percent Moisture	98.0	%	0.10	1		03/31/22 16:58		N2

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50312383

QC Batch: 670545

QC Batch Method: EPA 3050B

Analysis Method: EPA 6020

Analysis Description: 6020 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50312383001

METHOD BLANK: 3087997

Matrix: Solid

Associated Lab Samples: 50312383001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	04/12/22 14:03	
Cadmium	mg/kg	<0.050	0.050	04/12/22 14:03	
Chromium	mg/kg	<0.20	0.20	04/12/22 14:03	
Copper	mg/kg	<0.10	0.10	04/12/22 14:03	
Lead	mg/kg	<0.10	0.10	04/12/22 14:03	
Molybdenum	mg/kg	<0.10	0.10	04/12/22 14:03	N2
Nickel	mg/kg	<0.050	0.050	04/12/22 14:03	
Selenium	mg/kg	<0.10	0.10	04/12/22 14:03	
Silver	mg/kg	<0.050	0.050	04/12/22 14:03	
Zinc	mg/kg	<0.50	0.50	04/12/22 14:03	

LABORATORY CONTROL SAMPLE: 3087998

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.9	97	80-120	
Cadmium	mg/kg	4	4.0	99	80-120	
Chromium	mg/kg	4	4.2	105	80-120	
Copper	mg/kg	4	4.2	105	80-120	
Lead	mg/kg	4	4.3	108	80-120	
Molybdenum	mg/kg	4	4.2	106	80-120	N2
Nickel	mg/kg	4	4.1	101	80-120	
Selenium	mg/kg	4	3.9	98	80-120	
Silver	mg/kg	4	4.3	108	80-120	
Zinc	mg/kg	4	3.8	96	80-120	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50312383

QC Batch:	669002	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	Dry Weight/Percent Moisture
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50312383001

SAMPLE DUPLICATE: 3080492

Parameter	Units	50312475001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	5.1	5.3	3	5	N2

SAMPLE DUPLICATE: 3080493

Parameter	Units	50312550001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	4.2	10.1			N2,R1

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALIFIERS

Project: Biosolids
Pace Project No.: 50312383

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
R1	RPD value was outside control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 50312383

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50312383001	Biosolids Digester #4	EPA 3050B	670545	EPA 6020	670647
50312383001	Biosolids Digester #4	SM 2540G	669002		

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CHAIN-OF-CUSTODY / Analytical Request Document

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be completed. Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace Terms and Conditions found at <https://info.pacelabs.com/terms-conditions>

WO#: 50312383



50312383

Section A

Required Client Information:

Company: City of Cadillac
Address: 1121 Plett Rd
Cadillac, MI 49601
Email: ctomaszewski@cadillac-mi.net
Phone: 231-775-2368
Fax: [blank]
Requested Due Date: [blank]

Section B

Required Project Information:

Report To: Cindy Tomaszewski
Copy To: [blank]
Purchase Order #: [blank]
Project Name: Biosolids
Project #: [blank]

Section C

Invoice Information:

Attention: Cindy Tomaszewski
Company Name: [blank]
Address: [blank]
Pace Quote: [blank]
Pace Project Manager: brian.hall@pacelabs.com
Pace Profile #: 8311

Regulatory Agency

State / Location

MI

Requested Analysis Filtered (Y/N)

ITEM #	MATRIX	CODE	COLLECTED		SAMPLE TYPE (G=GRAB C=COMP)	MATRIX CODE (see valid codes to left)	RELINQUISHED BY / AFFILIATION		ACCEPTED BY / AFFILIATION		SAMPLE CONDITIONS		Received on	Ice (Y/N)	Custody (Y/N)	Sealed Cooler (Y/N)	Intact Samples (Y/N)
			START	END			DATE	TIME	DATE	TIME	DATE	TIME					
1	Biosolids Digester #4																
2																	
3																	
4																	
5																	
6																	
7																	
8																	
9																	
10																	
11																	
12																	
ADDITIONAL COMMENTS																	
TEMP in C																	
Requested Analysis Filtered (Y/N)																	
Analyses Test Y/N																	
Preservatives																	
H2SO4																	
HNO3																	
HCl																	
NaOH																	
Na2S2O3																	
Methanol																	
Other																	
# OF CONTAINERS																	
Unpreserved																	
Metals - 6020																	
Residual Chlorine (Y/N)																	

SAMPLER NAME AND SIGNATURE

PRINT Name of SAMPLER: TASHA LACTKO

SIGNATURE of SAMPLER: T. Lactko

DATE Signed: 3-28-22

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>3/29/1137</u>		Evaluated by: <u>DAS</u>		Affix Workorder/Label Here or WO#: 50312383 PM: BJH Due Date: 04/12/22 CLIENT: GR-Cadillac	
Client: <u>City of Cadillac</u>					
Project Manager: <u>BJH</u>		Profile ID: <u>8311</u>			
Rush TAT Requested: YES NO		Due Date:			
Lab Notified of Rush or Short Holds: YES NO		Non Conformance Form Required: YES NO			
Samples Received Via: FedEx UPS Client Pace Courier Other: _____					Comments:
Custody Seals Present and Intact:		YES	NO	NA	
Received Sample Information Form(s): Drinking Waters Only		YES	NO	NA	
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)		YES	NO	N/A	
Short Holds Present (< 72 Hours):		YES	NO		
Samples Received in Hold:		YES	NO		
Custody Signatures Present:		YES	NO		
Collector Signature Present:		YES	NO		
Packing Material Used:		YES	NO		
Samples Collected Today and On Ice:		YES	NO	N/A	
IR Gun #: 280 281		Digital Thermometer #: 282 283			
Ice Type: WET Bagged / WET Loose BLUE NONE		1. Cooler Temp Upon Receipt: <u>6.9/5.9</u> °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED		Temp should be 0-6°C (Initial/Corrected)			
Temp Blank Received:		YES	NO		
Containers Intact:		YES	NO		
Correct Containers:		YES	NO		
Sufficient Volume:		YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl		YES	NO	N/A	
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ (SVOC/Pest 625, PCB 608, Total/Amenable Cyanide)		YES	NO	N/A	
VOA Headspace Acceptable (<6mm):		YES	NO	N/A	
Trip Blank Received: HCl MeOH TSP OTHER		YES	NO		
Comments:		2. Cooler Temp Upon Receipt: _____ °C			
		3. Cooler Temp Upon Receipt: _____ °C			
		4. Cooler Temp Upon Receipt: _____ °C			